

Executive Summary

Made by: Siji Akinkugbe, Helen Chen

1. Introduction

This project develops a quantitative stock return prediction model using gradient-boosted regression trees (XGBoost). The objective is to predict one-month-ahead stock returns using a set of fundamental and price-based features, then construct long-short decile portfolios to evaluate out-of-sample performance. The sample covers U.S. equities listed on NYSE, AMEX, and Nasdaq from 1995 to 2024, drawn from CRSP and Compustat. Portfolio performance is evaluated out-of-sample from January 2000 onward.

2. Features and Rationale

The model uses 10 cross-sectionally percentile-ranked features (mapped to [0,1] each month). Eight are standard features from class: market capitalization, new share issuance (new_issue), asset growth (investment), accruals, book-to-market ratio (b2m), momentum (ret_2_12, compounded returns from month t-12 to t-2), cash flow to total assets (CashFlow2TA), and cash flow to price (CashFlow2Prc).

Two new features were added:

- **ROE (Return on Equity) = Income Before Extraordinary Items / Book Equity.** ROE measures how efficiently a firm converts shareholders' equity into profit. Unlike CashFlow2TA (which scales by assets) or CashFlow2Prc (which scales by market cap), ROE isolates the accrual-based profitability signal relative to book equity — a distinct dimension of firm quality documented in the Fama-French profitability factor (Novy-Marx 2013). Firms with persistently high ROE tend to be underpriced relative to their earnings power.
- **Short-Term Reversal = Prior Month Return (RET shifted by 1).** Stocks that experience extreme gains or losses in one month tend to partially reverse the following month, a robust anomaly documented by Jegadeesh (1990). This is a pure CRSP-based, high-frequency signal that is explicitly excluded from the momentum feature (ret_2_12 skips month t-1), making it a non-overlapping and complementary predictor.

3. Model and Implementation

The model is an XGBoost gradient-boosted regression tree (max depth = 4, learning rate = 0.3, subsample = 1.0, colsample_bytree = 0.8). The target variable is the cross-sectional demeaned monthly return (stock return minus the equal-weighted market return that month), which removes the common market component and focuses prediction on relative performance.

A rolling-window approach is used: for each prediction month t , the model is trained on the 60 months (5 years) immediately preceding t and predicts returns for month $t+1$. This walk-forward structure avoids look-ahead bias and simulates realistic out-of-sample deployment. Training runs from 1995 through 2024, with 290 rolling prediction months. The backtest evaluation period begins in January 2000 (covering 299 months out-of-sample).

4. Portfolio Construction and Performance

Each month, stocks are sorted into 10 equal-sized deciles based on predicted adjusted returns. Decile 0 contains the lowest-predicted-return stocks (short leg) and Decile 9 contains the highest (long leg). The "diff" portfolio is long Decile 9 and short Decile 0, rebalanced monthly with equal weighting within each decile.

Portfolio returns show a broadly monotonic pattern from Decile 0 to Decile 9, consistent with the model capturing a genuine return signal. Full performance metrics are reported in Table 1 below. Key highlights for the diff (long-short) portfolio: annualized return of 14.0%, annualized volatility of 18.9%, Sharpe ratio of 0.74, and a maximum drawdown of -33.9% over the evaluation period. The t -statistic on mean monthly returns is 3.71, indicating statistical significance.

5. Market Neutrality

To test market neutrality, the diff portfolio's monthly returns were regressed on the Fama-French market excess return (Mkt-RF). The estimated market beta is **-0.3281 ($p < 0.001$)**, and the regression R^2 is 7.6%. The portfolio is therefore **not market-neutral** — it carries a statistically significant negative market exposure. This arises because the short leg (Decile 0, low-predicted-return stocks) tends to include high-beta, distressed, or speculative names; shorting these creates a net short-market position. The portfolio's statistically significant alpha of **1.37% per month ($p < 0.001$)** confirms that the strategy generates substantial returns beyond its market exposure.

6. Handling Outliers

Outliers in financial data — such as extreme earnings spikes, near-zero book equity inflating ROE to infinity, or data errors in micro-cap stocks — can distort model inputs and bias predictions. The model addresses this through three mechanisms. First, all

features are converted to cross-sectional percentile ranks each month, mapping every value to $[0, 1]$ regardless of magnitude. A firm with an astronomically high raw ROE (e.g., due to near-zero equity) simply receives a rank near 1.0, with no outsized numerical influence. Second, infinite and undefined values (resulting from division by zero in ROE or b2m) are explicitly replaced with NaN and dropped before model training. Third, stocks with market capitalization below \$10 million are excluded — micro-caps are disproportionately subject to data errors, extreme illiquidity, and return anomalies unrelated to genuine predictive signals. Beyond these pre-processing steps, XGBoost's tree-based structure is inherently more robust to outliers than linear models, since splits are based on rank order within a node rather than the magnitude of individual values.

7. Key Takeaways

The XGBoost regression tree model successfully identifies a cross-sectional return signal, producing a long-short portfolio with a Sharpe ratio of 0.74 and statistically significant alpha over the 2000–2024 evaluation period. The monotonic pattern of decile returns validates the model's predictive power. The two new features — ROE and short-term reversal — contribute distinct dimensions of profitability and price-reversal information not captured by the class feature set. The main limitation is that the diff portfolio carries a meaningful negative market beta (-0.33), meaning it is not fully market-neutral and can underperform sharply in strong bull markets (consistent with the maximum drawdown of -33.9%). Future improvements could include adding a market-beta neutralization step, incorporating additional factors (e.g., earnings surprise, analyst revision), or using a longer training window with regularization tuning.

Tables

Table 1: Decile Portfolio and Diff Portfolio Performance Metrics (Evaluation Period: Jan 2000 – Dec 2024, 299 months)

Decile	Mean Monthly Ret	Std Dev	t-stat	Ann. Return	Ann. Volatility	Sharpe Ratio
0 (Short)	0.17%	9.34%	0.31	2.0%	32.3%	0.06
1	0.72%	7.28%	1.71	8.6%	25.2%	0.34
2	0.90%	6.40%	2.42	10.8%	22.2%	0.48
3	1.02%	5.72%	3.09	12.3%	19.8%	0.62
4	1.08%	5.48%	3.39	12.9%	19.0%	0.68
5	1.03%	5.46%	3.26	12.4%	18.9%	0.65
6	1.08%	5.51%	3.39	13.0%	19.1%	0.68
7	1.11%	5.62%	3.42	13.4%	19.5%	0.69
8	1.20%	6.24%	3.33	14.5%	21.6%	0.67
9 (Long)	1.34%	7.78%	2.97	16.1%	26.9%	0.60
Diff (9–0)	1.17%	5.44%	3.71	14.0%	18.9%	0.74

Diff Portfolio Maximum Drawdown: -33.9%

Table 2: Market Neutrality Test — OLS Regression of Diff Portfolio on Mkt-RF (299 months)

	Coefficient	Std. Error	t-stat	p-value
Alpha (const)	0.0137	0.003	4.48	<0.001
Market Beta (Mkt-RF)	-0.3281	0.066	-4.95	<0.001
R-squared	0.076			

Table 3: Feature List and Formulas

Feature	Formula	Source
marketcap	$\text{SHROUT} \times \text{PRC}$ (lagged 1 month)	CRSP
new_issue	$(\text{SHROUT_adj} - \text{SHROUT_adj_lag12}) / \text{SHROUT_adj_lag12}$	CRSP
investment	$\text{AT} / \text{AT_lag} - 1$	Compustat
accruals	$(\text{IB} - \text{OANCF}) / \text{AT}$	Compustat
b2m	$\text{CEQ} / \text{marketcap}$	Compustat / CRSP
ret_2_12	$\prod (1 + \text{RET})$ for months t-12 to t-2	CRSP
CashFlow2TA	OANCF / AT	Compustat
CashFlow2Prc	$\text{OANCF} / \text{marketcap}$	Compustat / CRSP
ROE (new)	IB / CEQ	Compustat
short_term_reversal (new)	RET(t-1)	CRSP

All features are cross-sectionally percentile-ranked within each month prior to model input.